```
> restart: unprotect('D'):

m := 3;

twist := - 6;

d := 15;                    degre := d + 1:

##

#SS  := x^(d+1) + 1^(d+1) + 0*x^6:
#HSS := X^(d+1) + T^(d+1) + 0*X^6:

SS  := x^(d+1) + 1^(d+1) + 1*x^6:
HSS := X^(d+1) + T^(d+1) + 1*T^(d+1-6)*X^6:

##

Jet_xR := expand(

+ (1/Ry^(m-0*2)) * add(AA||j * x1^(m-0*3-j) * (R1/RR)^j *
Delta_xR^0, j=0..m-0*3)

+ (1/Ry^(m-1*2)) * add(BB||j * x1^(m-1*3-j) * (R1/RR)^j *
Delta_xR^1, j=0..m-1*3)

+ (1/Ry^(m-2*2)) * add(CC||j * x1^(m-2*3-j) * (R1/RR)^j *
Delta_xR^2, j=0..m-2*3)

+ (1/Ry^(m-3*2)) * add(DD||j * x1^(m-3*3-j) * (R1/RR)^j *
Delta_xR^3, j=0..m-3*3)

+ (1/Ry^(m-4*2)) * add(EE||j * x1^(m-4*3-j) * (R1/RR)^j *
Delta_xR^4, j=0..m-4*3) ):

##

for l from 0 to m - 0*3 do dA[l] := (m-0*2)*d - m + l*1 + 0*1 +
twist: od:

for l from 0 to m - 1*3 do dB[l] := (m-1*2)*d - m + l*1 + 1*1 +
twist: od:

for l from 0 to m - 2*3 do dC[l] := (m-2*2)*d - m + l*1 + 2*1 +
twist: od:

for l from 0 to m - 3*3 do dD[l] := (m-3*2)*d - m + l*1 + 3*1 +
twist: od:

for l from 0 to m - 4*3 do dE[l] := (m-4*2)*d - m + l*1 + 4*1 +
twist: od:

##
```

```
Delta_xR := x1 * (R2/RR - R1*R1/RR^2) - x2 * (R1/RR):

Solx1 := - y1*Ry/Rx + R1/Rx:

Solx2 := - Rxx*y1^2*Ry^2/Rx^3 + 2*Rxx*y1*Ry*R1/Rx^3 - Rxx*
R1^2/Rx^3 + 2*y1^2*Rxy*Ry/Rx^2
      - 2*y1*Rxy*R1/Rx^2 - y1^2*Ryy/Rx - y2*Ry/Rx + R2/Rx:

##

Jet_Ry := sort(mtaylor(

                  expand(subs(R2=-(RR^2*Delta_Ry-RR*R1*y2-
R1^2*y1)/(RR*y1),
                    expand(subs({x1=Solx1,x2=Solx2},
                Jet_xR)))),

       [R1,y1,Delta_Ry], 100), [R1,y1,Delta_Ry]):

##

printlevel := 2:
for k from 0 to m/3  do
for j from 0 to m-3*k do

EEq[m-j-3*k,j,k] := factor(

 Ry^(m-j-3*k)*RR^(m-j-3*k)*Rx^(m-2*k)
*
 coeftayl(Jet_Ry, [R1,y1,Delta_Ry]=[0,0,0], [m-j-3*k,j,k]) ):

od: od:

## EQUATION RR EXPLICITE

RR  := y^(d+1) + SS;
HRR := Y^(d+1) + HSS;

RY := diff(HRR, Y):

Rx := diff(RR,x):
Ry := diff(RR,y):

Rxx := diff(RR,x,x):
Rxy := diff(RR,x,y):
Ryy := diff(RR,y,y):

##

printlevel := 2:
for k from 0 to m/3  do
for j from 0 to m-3*k do

Eq[m-j-3*k,j,k] := mtaylor(EEq[m-j-3*k,j,k], y, (m-j-3*k)*d):

Var[m-j-3*k,j,k] := indets(Eq[m-j-3*k,j,k]) minus {x,y}:
```

```
od: od:
##
```

$$m := 3$$
$$twist := -6$$
$$d := 15$$
$$RR := x^{16} + y^{16} + x^{6} + 1$$
$$HRR := T^{16} + T^{10} X^{6} + X^{16} + Y^{16} \quad (1)$$

```
> ## EXPAND ALL UNKNOWNS UP TO y^(1*d)

for l from 0 to m - 0*3 do
AA||l := add(add(A||l[i,j]*x^i*y^j, i=0..dA[l]-j), j=0..min(dA
[l],1*d-1)): od:

for l from 0 to m - 1*3 do
BB||l := add(add(B||l[i,j]*x^i*y^j, i=0..dB[l]-j), j=0..min(dB
[l],1*d-1)): od:

for l from 0 to m - 2*3 do
CC||l := add(add(C||l[i,j]*x^i*y^j, i=0..dC[l]-j), j=0..min(dC
[l],1*d-1)): od:

for l from 0 to m - 3*3 do
DD||l := add(add(D||l[i,j]*x^i*y^j, i=0..dD[l]-j), j=0..min(dD
[l],1*d-1)): od:

for l from 0 to m - 4*3 do
EE||l := add(add(E||l[i,j]*x^i*y^j, i=0..dE[l]-j), j=0..min(dE
[l],1*d-1)): od:

## SYSTEM FOR DIVISIBILITY BY y^(1*d)

Systemey1d := {}:

for k from 0 to (m-1)/3   do

Coeffs_y1d[1,m-1-3*k,k] := {coeffs(expand(mtaylor(Eq[1,m-1-3*k,
k], y, 1*d)), [x,y])}:

Systemey1d := Systemey1d union Coeffs_y1d[1,m-1-3*k,k]:

od:

Systeme_y1d := Systemey1d:

## VARIABLES FOR DIVISIBILITY BY y^(1*d)

Variables_y1d := indets(Systeme_y1d):
```

```
## RESOLUTION y^(1*d)

Solutions_y1d := solve(Systeme_y1d, Variables_y1d):

assign(Solutions_y1d):

##

for l from 0 to m - 0*3 do nops(expand(AA||l)) od;
for l from 0 to m - 1*3 do nops(expand(BB||l)) od;
for l from 0 to m - 2*3 do nops(expand(CC||l)) od;
for l from 0 to m - 3*3 do nops(expand(DD||l)) od;
for l from 0 to m - 4*3 do nops(expand(EE||l)) od;
```

$$l := 0$$
$$845$$
$$l := 1$$
$$1125$$
$$l := 2$$
$$480$$
$$l := 3$$
$$495$$
$$l := 4$$
$$l := 0$$
$$39$$
$$l := 1$$
$$l := 0$$
$$l := 0$$
$$l := 0 \tag{2}$$

```
> ## EXPAND ALL UNKNOWNS UP TO y^(2*d)

for l from 0 to m - 0*3 do
AA||l := add(add(A||l[i,j]*x^i*y^j, i=0..dA[l]-j), j=0..min(dA
[l],2*d-1)): od:

for l from 0 to m - 1*3 do
BB||l := add(add(B||l[i,j]*x^i*y^j, i=0..dB[l]-j), j=0..min(dB
[l],2*d-1)): od:

for l from 0 to m - 2*3 do
CC||l := add(add(C||l[i,j]*x^i*y^j, i=0..dC[l]-j), j=0..min(dC
[l],2*d-1)): od:
```

```
for l from 0 to m - 3*3 do
DD||l := add(add(D||l[i,j]*x^i*y^j, i=0..dD[l]-j), j=0..min(dD
[l],2*d-1)): od:

for l from 0 to m - 4*3 do
EE||l := add(add(E||l[i,j]*x^i*y^j, i=0..dE[l]-j), j=0..min(dE
[l],2*d-1)): od:

## SYSTEM FOR DIVISIBILITY BY y^(2*d)

Systemey2d := {}:

for k from 0 to (m-2)/3   do

Coeffs_y2d[2,m-2-3*k,k] := {coeffs(expand(mtaylor(Eq[2,m-2-3*k,
k], y, 2*d)), [x,y])}:

Systemey2d := Systemey2d union Coeffs_y2d[2,m-2-3*k,k]:

od:

Systeme_y2d := Systemey2d:

## VARIABLES FOR DIVISIBILITY BY y^(2*d)

Variables_y2d := indets(Systeme_y2d):

## RESOLUTION y^(2*d)

Solutions_y2d := solve(Systeme_y2d, Variables_y2d):

assign(Solutions_y2d):

##

for l from 0 to m - 0*3 do nops(expand(AA||l)) od;
for l from 0 to m - 1*3 do nops(expand(BB||l)) od;
for l from 0 to m - 2*3 do nops(expand(CC||l)) od;
for l from 0 to m - 3*3 do nops(expand(DD||l)) od;
for l from 0 to m - 4*3 do nops(expand(EE||l)) od;
```

$$l := 0$$
$$149$$
$$l := 1$$
$$292$$
$$l := 2$$
$$403$$
$$l := 3$$
$$765$$
$$l := 4$$
$$l := 0$$

$$36$$
$$l := 1$$
$$l := 0$$
$$l := 0$$
$$l := 0 \tag{3}$$

```
> ## EXPAND ALL UNKNOWNS UP TO y^(3*d)

for l from 0 to m - 0*3 do
AA||l := add(add(A||l[i,j]*x^i*y^j, i=0..dA[l]-j), j=0..min(dA
[l],3*d-1)): od:

for l from 0 to m - 1*3 do
BB||l := add(add(B||l[i,j]*x^i*y^j, i=0..dB[l]-j), j=0..min(dB
[l],3*d-1)): od:

for l from 0 to m - 2*3 do
CC||l := add(add(C||l[i,j]*x^i*y^j, i=0..dC[l]-j), j=0..min(dC
[l],3*d-1)): od:

for l from 0 to m - 3*3 do
DD||l := add(add(D||l[i,j]*x^i*y^j, i=0..dD[l]-j), j=0..min(dD
[l],3*d-1)): od:

for l from 0 to m - 4*3 do
EE||l := add(add(E||l[i,j]*x^i*y^j, i=0..dE[l]-j), j=0..min(dE
[l],3*d-1)): od:

## SYSTEM FOR DIVISIBILITY BY y^(3*d)

Systemey3d := {}:

for k from 0 to (m-3)/3   do

Coeffs_y3d[3,m-3-3*k,k] := {coeffs(expand(mtaylor(Eq[3,m-3-3*k,
k], y, 3*d)), [x,y])}:

Systemey3d := Systemey3d union Coeffs_y3d[3,m-3-3*k,k]:

od:

Systeme_y3d := Systemey3d:

## VARIABLES FOR DIVISIBILITY BY y^(3*d)

Variables_y3d := indets(Systeme_y3d):

## RESOLUTION y^(3*d)

Solutions_y3d := solve(Systeme_y3d, Variables_y3d):
```

```
assign(Solutions_y3d):

##

for l from 0 to m - 0*3 do nops(expand(AA||l)) od;
for l from 0 to m - 1*3 do nops(expand(BB||l)) od;
for l from 0 to m - 2*3 do nops(expand(CC||l)) od;
for l from 0 to m - 3*3 do nops(expand(DD||l)) od;
for l from 0 to m - 4*3 do nops(expand(EE||l)) od;
```

$$l := 0$$
$$\mathbf{1}$$
$$l := 1$$
$$\mathbf{1}$$
$$l := 2$$
$$\mathbf{1}$$
$$l := 3$$
$$\mathbf{1}$$
$$l := 4$$
$$l := 0$$
$$\mathbf{1}$$
$$l := 1$$
$$l := 0$$
$$l := 0$$
$$l := 0 \tag{4}$$

```
> ## EXPAND ALL UNKNOWNS UP TO y^(4*d)

for l from 0 to m - 0*3 do
AA||l := add(add(A||l[i,j]*x^i*y^j, i=0..dA[l]-j), j=0..min(dA
[l],4*d-1)): od:

for l from 0 to m - 1*3 do
BB||l := add(add(B||l[i,j]*x^i*y^j, i=0..dB[l]-j), j=0..min(dB
[l],4*d-1)): od:

for l from 0 to m - 2*3 do
CC||l := add(add(C||l[i,j]*x^i*y^j, i=0..dC[l]-j), j=0..min(dC
[l],4*d-1)): od:

for l from 0 to m - 3*3 do
DD||l := add(add(D||l[i,j]*x^i*y^j, i=0..dD[l]-j), j=0..min(dD
[l],4*d-1)): od:

for l from 0 to m - 4*3 do
```

```
EE||l := add(add(E||l[i,j]*x^i*y^j, i=0..dE[l]-j), j=0..min(dE
[l],4*d-1)): od:

## SYSTEM FOR DIVISIBILITY BY y^(4*d)

Systemey4d := {}:

for k from 0 to (m-4)/3   do

Coeffs_y4d[4,m-4-3*k,k] := {coeffs(expand(mtaylor(Eq[4,m-4-3*k,
k], y, 4*d)), [x,y])}:

Systemey4d := Systemey4d union Coeffs_y4d[4,m-4-3*k,k]:

od:

Systeme_y4d := Systemey4d:

## VARIABLES FOR DIVISIBILITY BY y^(4*d)

Variables_y4d := indets(Systeme_y4d):

## RESOLUTION y^(4*d)

Solutions_y4d := solve(Systeme_y4d, Variables_y4d):

assign(Solutions_y4d):

##

for l from 0 to m - 0*3 do nops(expand(AA||l)) od;
for l from 0 to m - 1*3 do nops(expand(BB||l)) od;
for l from 0 to m - 2*3 do nops(expand(CC||l)) od;
for l from 0 to m - 3*3 do nops(expand(DD||l)) od;
for l from 0 to m - 4*3 do nops(expand(EE||l)) od;
```

$$l := 0$$
$$\mathbf{1}$$
$$l := 1$$
$$\mathbf{1}$$
$$l := 2$$
$$\mathbf{1}$$
$$l := 3$$
$$\mathbf{1}$$
$$l := 4$$
$$l := 0$$
$$\mathbf{1}$$
$$l := 1$$
$$l := 0$$
$$l := 0$$

$$l := 0 \tag{5}$$

```
> ## EXPAND ALL UNKNOWNS UP TO y^(5*d)

for l from 0 to m - 0*3 do
AA||l := add(add(A||l[i,j]*x^i*y^j, i=0..dA[l]-j), j=0..min(dA
[l],5*d-1)): od:

for l from 0 to m - 1*3 do
BB||l := add(add(B||l[i,j]*x^i*y^j, i=0..dB[l]-j), j=0..min(dB
[l],5*d-1)): od:

for l from 0 to m - 2*3 do
CC||l := add(add(C||l[i,j]*x^i*y^j, i=0..dC[l]-j), j=0..min(dC
[l],5*d-1)): od:

for l from 0 to m - 3*3 do
DD||l := add(add(D||l[i,j]*x^i*y^j, i=0..dD[l]-j), j=0..min(dD
[l],5*d-1)): od:

for l from 0 to m - 4*3 do
EE||l := add(add(E||l[i,j]*x^i*y^j, i=0..dE[l]-j), j=0..min(dE
[l],5*d-1)): od:

## SYSTEM FOR DIVISIBILITY BY y^(5*d)

Systemey5d := {}:

for k from 0 to (m-5)/3   do

Coeffs_y5d[5,m-5-3*k,k] := {coeffs(expand(mtaylor(Eq[5,m-5-3*k,
k], y, 5*d)), [x,y])}:

Systemey5d := Systemey5d union Coeffs_y5d[5,m-5-3*k,k]:

od:

Systeme_y5d := Systemey5d:

## VARIABLES FOR DIVISIBILITY BY y^(5*d)

Variables_y5d := indets(Systeme_y5d):

## RESOLUTION y^(5*d)

Solutions_y5d := solve(Systeme_y5d, Variables_y5d):

assign(Solutions_y5d):

##

for l from 0 to m - 0*3 do nops(expand(AA||l)) od;
for l from 0 to m - 1*3 do nops(expand(BB||l)) od;
```

```
for l from 0 to m - 2*3 do nops(expand(CC||l)) od;
for l from 0 to m - 3*3 do nops(expand(DD||l)) od;
for l from 0 to m - 4*3 do nops(expand(EE||l)) od;
```

$$l := 0$$
$$\mathbf{1}$$
$$l := 1$$
$$\mathbf{1}$$
$$l := 2$$
$$\mathbf{1}$$
$$l := 3$$
$$\mathbf{1}$$
$$l := 4$$
$$l := 0$$
$$\mathbf{1}$$
$$l := 1$$
$$l := 0$$
$$l := 0$$
$$l := 0 \qquad (6)$$

```
> ## EXPAND ALL UNKNOWNS UP TO y^(6*d)

for l from 0 to m - 0*3 do
AA||l := add(add(A||l[i,j]*x^i*y^j, i=0..dA[l]-j), j=0..min(dA
[l],6*d-1)): od:

for l from 0 to m - 1*3 do
BB||l := add(add(B||l[i,j]*x^i*y^j, i=0..dB[l]-j), j=0..min(dB
[l],6*d-1)): od:

for l from 0 to m - 2*3 do
CC||l := add(add(C||l[i,j]*x^i*y^j, i=0..dC[l]-j), j=0..min(dC
[l],6*d-1)): od:

for l from 0 to m - 3*3 do
DD||l := add(add(D||l[i,j]*x^i*y^j, i=0..dD[l]-j), j=0..min(dD
[l],6*d-1)): od:

for l from 0 to m - 4*3 do
EE||l := add(add(E||l[i,j]*x^i*y^j, i=0..dE[l]-j), j=0..min(dE
[l],6*d-1)): od:

## SYSTEM FOR DIVISIBILITY BY y^(6*d)
```

```
Systemey6d := {}:

for k from 0 to (m-6)/3   do

Coeffs_y6d[6,m-6-3*k,k] := {coeffs(expand(mtaylor(Eq[6,m-6-3*k,
k], y, 6*d)), [x,y])}:

Systemey6d := Systemey6d union Coeffs_y6d[6,m-6-3*k,k]:

od:

Systeme_y6d := Systemey6d:

## VARIABLES FOR DIVISIBILITY BY y^(6*d)

Variables_y6d := indets(Systeme_y6d):

## RESOLUTION y^(6*d)

Solutions_y6d := solve(Systeme_y6d, Variables_y6d):

assign(Solutions_y6d):

##

for l from 0 to m - 0*3 do nops(expand(AA||l)) od;
for l from 0 to m - 1*3 do nops(expand(BB||l)) od;
for l from 0 to m - 2*3 do nops(expand(CC||l)) od;
for l from 0 to m - 3*3 do nops(expand(DD||l)) od;
for l from 0 to m - 4*3 do nops(expand(EE||l)) od;
```

$$l := 0$$
$$\mathbf{1}$$
$$l := 1$$
$$\mathbf{1}$$
$$l := 2$$
$$\mathbf{1}$$
$$l := 3$$
$$\mathbf{1}$$
$$l := 4$$
$$l := 0$$
$$\mathbf{1}$$
$$l := 1$$
$$l := 0$$
$$l := 0$$
$$l := 0 \tag{7}$$

```
> ## EXPAND ALL UNKNOWNS UP TO y^(7*d)

  for l from 0 to m - 0*3 do
  AA||l := add(add(A||l[i,j]*x^i*y^j, i=0..dA[l]-j), j=0..min(dA
  [l],7*d-1)): od:

  for l from 0 to m - 1*3 do
  BB||l := add(add(B||l[i,j]*x^i*y^j, i=0..dB[l]-j), j=0..min(dB
  [l],7*d-1)): od:

  for l from 0 to m - 2*3 do
  CC||l := add(add(C||l[i,j]*x^i*y^j, i=0..dC[l]-j), j=0..min(dC
  [l],7*d-1)): od:

  for l from 0 to m - 3*3 do
  DD||l := add(add(D||l[i,j]*x^i*y^j, i=0..dD[l]-j), j=0..min(dD
  [l],7*d-1)): od:

  for l from 0 to m - 4*3 do
  EE||l := add(add(E||l[i,j]*x^i*y^j, i=0..dE[l]-j), j=0..min(dE
  [l],7*d-1)): od:

  ## SYSTEM FOR DIVISIBILITY BY y^(7*d)

  Systemey7d := {}:

  for k from 0 to (m-7)/3   do

  Coeffs_y7d[7,m-7-3*k,k] := {coeffs(expand(mtaylor(Eq[7,m-7-3*k,
  k], y, 7*d)), [x,y])}:

  Systemey7d := Systemey7d union Coeffs_y7d[7,m-7-3*k,k]:

  od:

  Systeme_y7d := Systemey7d:

  ## VARIABLES FOR DIVISIBILITY BY y^(7*d)

  Variables_y7d := indets(Systeme_y7d):

  ## RESOLUTION y^(7*d)

  Solutions_y7d := solve(Systeme_y7d, Variables_y7d):

  assign(Solutions_y7d):

  ##

  for l from 0 to m - 0*3 do nops(expand(AA||l)) od;
  for l from 0 to m - 1*3 do nops(expand(BB||l)) od;
  for l from 0 to m - 2*3 do nops(expand(CC||l)) od;
  for l from 0 to m - 3*3 do nops(expand(DD||l)) od;
  for l from 0 to m - 4*3 do nops(expand(EE||l)) od;
```

$$l := 0$$

$$\mathbf{1}$$
$$l := 1$$
$$\mathbf{1}$$
$$l := 2$$
$$\mathbf{1}$$
$$l := 3$$
$$\mathbf{1}$$
$$l := 4$$
$$l := 0$$
$$\mathbf{1}$$
$$l := 1$$
$$l := 0$$
$$l := 0$$
$$l := 0 \tag{8}$$

```
>
## EXPAND ALL UNKNOWNS UP TO y^(8*d)

for l from 0 to m - 0*3 do
AA||l := add(add(A||l[i,j]*x^i*y^j, i=0..dA[l]-j), j=0..min(dA[l],8*d-1)): od:

for l from 0 to m - 1*3 do
BB||l := add(add(B||l[i,j]*x^i*y^j, i=0..dB[l]-j), j=0..min(dB[l],8*d-1)): od:

for l from 0 to m - 2*3 do
CC||l := add(add(C||l[i,j]*x^i*y^j, i=0..dC[l]-j), j=0..min(dC[l],8*d-1)): od:

for l from 0 to m - 3*3 do
DD||l := add(add(D||l[i,j]*x^i*y^j, i=0..dD[l]-j), j=0..min(dD[l],8*d-1)): od:

for l from 0 to m - 4*3 do
EE||l := add(add(E||l[i,j]*x^i*y^j, i=0..dE[l]-j), j=0..min(dE[l],8*d-1)): od:

## SYSTEM FOR DIVISIBILITY BY y^(8*d)

Systemey8d := {}:

for k from 0 to (m-8)/3  do

Coeffs_y8d[8,m-8-3*k,k] := {coeffs(expand(mtaylor(Eq[8,m-8-3*k,k], y, 8*d)), [x,y])}:
```

```
Systemey8d := Systemey8d union Coeffs_y8d[8,m-8-3*k,k]:
od:
Systeme_y8d := Systemey8d:
## VARIABLES FOR DIVISIBILITY BY y^(8*d)
Variables_y8d := indets(Systeme_y8d):
## RESOLUTION y^(8*d)
Solutions_y8d := solve(Systeme_y8d, Variables_y8d):
assign(Solutions_y8d):
##
for l from 0 to m - 0*3 do nops(expand(AA||l)) od;
for l from 0 to m - 1*3 do nops(expand(BB||l)) od;
for l from 0 to m - 2*3 do nops(expand(CC||l)) od;
for l from 0 to m - 3*3 do nops(expand(DD||l)) od;
for l from 0 to m - 4*3 do nops(expand(EE||l)) od;
```

$$l := 0$$
$$\mathbf{1}$$
$$l := 1$$
$$\mathbf{1}$$
$$l := 2$$
$$\mathbf{1}$$
$$l := 3$$
$$\mathbf{1}$$
$$l := 4$$
$$l := 0$$
$$\mathbf{1}$$
$$l := 1$$
$$l := 0$$
$$l := 0$$
$$l := 0 \tag{9}$$

```
> ## EXPAND ALL UNKNOWNS UP TO y^(9*d)
for l from 0 to m - 0*3 do
AA||l := add(add(A||l[i,j]*x^i*y^j, i=0..dA[l]-j), j=0..min(dA
[l],9*d-1)): od:
```

```
for l from 0 to m - 1*3 do
BB||l := add(add(B||l[i,j]*x^i*y^j, i=0..dB[l]-j), j=0..min(dB
[l],9*d-1)): od:

for l from 0 to m - 2*3 do
CC||l := add(add(C||l[i,j]*x^i*y^j, i=0..dC[l]-j), j=0..min(dC
[l],9*d-1)): od:

for l from 0 to m - 3*3 do
DD||l := add(add(D||l[i,j]*x^i*y^j, i=0..dD[l]-j), j=0..min(dD
[l],9*d-1)): od:

for l from 0 to m - 4*3 do
EE||l := add(add(E||l[i,j]*x^i*y^j, i=0..dE[l]-j), j=0..min(dE
[l],9*d-1)): od:

## SYSTEM FOR DIVISIBILITY BY y^(9*d)

Systemey9d := {}:

for k from 0 to (m-9)/3   do

Coeffs_y9d[9,m-9-3*k,k] := {coeffs(expand(mtaylor(Eq[9,m-9-3*k,
k], y, 9*d)), [x,y])}:

Systemey9d := Systemey9d union Coeffs_y9d[9,m-9-3*k,k]:

od:

Systeme_y9d := Systemey9d:

## VARIABLES FOR DIVISIBILITY BY y^(9*d)

Variables_y9d := indets(Systeme_y9d):

## RESOLUTION y^(9*d)

Solutions_y9d := solve(Systeme_y9d, Variables_y9d):

assign(Solutions_y9d):

##

for l from 0 to m - 0*3 do nops(expand(AA||l)) od;
for l from 0 to m - 1*3 do nops(expand(BB||l)) od;
for l from 0 to m - 2*3 do nops(expand(CC||l)) od;
for l from 0 to m - 3*3 do nops(expand(DD||l)) od;
for l from 0 to m - 4*3 do nops(expand(EE||l)) od;
```

$$l := 0$$

$$\mathbf{1}$$

$$l := 1$$

$$\mathbf{1}$$

$$l := 2$$

$$\mathbf{1}$$
$$l := 3$$
$$\mathbf{1}$$
$$l := 4$$
$$l := 0$$
$$\mathbf{1}$$
$$l := 1$$
$$l := 0$$
$$l := 0$$
$$l := 0 \tag{10}$$

```
>
## EXPAND ALL UNKNOWNS UP TO y^(10*d)

for l from 0 to m - 0*3 do
AA||l := add(add(A||l[i,j]*x^i*y^j, i=0..dA[l]-j), j=0..min(dA
[l],10*d-1)): od:

for l from 0 to m - 1*3 do
BB||l := add(add(B||l[i,j]*x^i*y^j, i=0..dB[l]-j), j=0..min(dB
[l],10*d-1)): od:

for l from 0 to m - 2*3 do
CC||l := add(add(C||l[i,j]*x^i*y^j, i=0..dC[l]-j), j=0..min(dC
[l],10*d-1)): od:

for l from 0 to m - 3*3 do
DD||l := add(add(D||l[i,j]*x^i*y^j, i=0..dD[l]-j), j=0..min(dD
[l],10*d-1)): od:

for l from 0 to m - 4*3 do
EE||l := add(add(E||l[i,j]*x^i*y^j, i=0..dE[l]-j), j=0..min(dE
[l],10*d-1)): od:

## SYSTEM FOR DIVISIBILITY BY y^(10*d)

Systemey10d := {}:

for k from 0 to (m-10)/3   do

Coeffs_y10d[10,m-10-3*k,k] := {coeffs(expand(mtaylor(Eq[10,m-10
-3*k,k], y, 10*d)), [x,y])}:

Systemey10d := Systemey10d union Coeffs_y10d[10,m-10-3*k,k]:

od:

Systeme_y10d := Systemey10d:
```

```
## VARIABLES FOR DIVISIBILITY BY y^(10*d)

Variables_y10d := indets(Systeme_y10d):

## RESOLUTION y^(10*d)

Solutions_y10d := solve(Systeme_y10d, Variables_y10d):

assign(Solutions_y10d):

##

for l from 0 to m - 0*3 do nops(expand(AA||l)) od;
for l from 0 to m - 1*3 do nops(expand(BB||l)) od;
for l from 0 to m - 2*3 do nops(expand(CC||l)) od;
for l from 0 to m - 3*3 do nops(expand(DD||l)) od;
for l from 0 to m - 4*3 do nops(expand(EE||l)) od;
```

$$l := 0$$
$$\mathbf{1}$$
$$l := 1$$
$$\mathbf{1}$$
$$l := 2$$
$$\mathbf{1}$$
$$l := 3$$
$$\mathbf{1}$$
$$l := 4$$
$$l := 0$$
$$\mathbf{1}$$
$$l := 1$$
$$l := 0$$
$$l := 0$$
$$l := 0 \tag{11}$$

```
>
## EXPAND ALL UNKNOWNS UP TO y^(11*d)

for l from 0 to m - 0*3 do
AA||l := add(add(A||l[i,j]*x^i*y^j, i=0..dA[l]-j), j=0..min(dA
[l],11*d-1)): od:

for l from 0 to m - 1*3 do
BB||l := add(add(B||l[i,j]*x^i*y^j, i=0..dB[l]-j), j=0..min(dB
[l],11*d-1)): od:

for l from 0 to m - 2*3 do
```

```
CC||l := add(add(C||l[i,j]*x^i*y^j, i=0..dC[l]-j), j=0..min(dC
[l],11*d-1)): od:

for l from 0 to m - 3*3 do
DD||l := add(add(D||l[i,j]*x^i*y^j, i=0..dD[l]-j), j=0..min(dD
[l],11*d-1)): od:

for l from 0 to m - 4*3 do
EE||l := add(add(E||l[i,j]*x^i*y^j, i=0..dE[l]-j), j=0..min(dE
[l],11*d-1)): od:

## SYSTEM FOR DIVISIBILITY BY y^(11*d)

Systemey11d := {}:

for k from 0 to (m-11)/3   do

Coeffs_y11d[11,m-11-3*k,k] := {coeffs(expand(mtaylor(Eq[11,m-11
-3*k,k], y, 11*d)), [x,y])}:

Systemey11d := Systemey11d union Coeffs_y11d[11,m-11-3*k,k]:

od:

Systeme_y11d := Systemey11d:

## VARIABLES FOR DIVISIBILITY BY y^(11*d)

Variables_y11d := indets(Systeme_y11d):

## RESOLUTION y^(11*d)

Solutions_y11d := solve(Systeme_y11d, Variables_y11d):

assign(Solutions_y11d):

##

for l from 0 to m - 0*3 do nops(expand(AA||l)) od;
for l from 0 to m - 1*3 do nops(expand(BB||l)) od;
for l from 0 to m - 2*3 do nops(expand(CC||l)) od;
for l from 0 to m - 3*3 do nops(expand(DD||l)) od;
for l from 0 to m - 4*3 do nops(expand(EE||l)) od;
```

$$l := 0$$

$$\mathbf{1}$$

$$l := 1$$

$$\mathbf{1}$$

$$l := 2$$

$$\mathbf{1}$$

$$l := 3$$

$$\mathbf{1}$$

$$l := 4$$
$$l := 0$$
$$\mathbf{1}$$
$$l := 1$$
$$l := 0$$
$$l := 0$$
$$l := 0 \tag{12}$$

```
> ## EXPAND ALL UNKNOWNS UP TO y^(12*d)

for l from 0 to m - 0*3 do
AA||l := add(add(A||l[i,j]*x^i*y^j, i=0..dA[l]-j), j=0..min(dA
[l],12*d-1)): od:

for l from 0 to m - 1*3 do
BB||l := add(add(B||l[i,j]*x^i*y^j, i=0..dB[l]-j), j=0..min(dB
[l],12*d-1)): od:

for l from 0 to m - 2*3 do
CC||l := add(add(C||l[i,j]*x^i*y^j, i=0..dC[l]-j), j=0..min(dC
[l],12*d-1)): od:

for l from 0 to m - 3*3 do
DD||l := add(add(D||l[i,j]*x^i*y^j, i=0..dD[l]-j), j=0..min(dD
[l],12*d-1)): od:

for l from 0 to m - 4*3 do
EE||l := add(add(E||l[i,j]*x^i*y^j, i=0..dE[l]-j), j=0..min(dE
[l],12*d-1)): od:

## SYSTEM FOR DIVISIBILITY BY y^(12*d)

Systemey12d := {}:

for k from 0 to (m-12)/3   do

Coeffs_y12d[12,m-12-3*k,k] := {coeffs(expand(mtaylor(Eq[12,m-12
-3*k,k], y, 12*d)), [x,y])}:

Systemey12d := Systemey12d union Coeffs_y12d[12,m-12-3*k,k]:

od:

Systeme_y12d := Systemey12d:

## VARIABLES FOR DIVISIBILITY BY y^(12*d)

Variables_y12d := indets(Systeme_y12d):

## RESOLUTION y^(12*d)
```

```
Solutions_y12d := solve(Systeme_y12d, Variables_y12d):

assign(Solutions_y12d):

##

for l from 0 to m - 0*3 do nops(expand(AA||l)) od;
for l from 0 to m - 1*3 do nops(expand(BB||l)) od;
for l from 0 to m - 2*3 do nops(expand(CC||l)) od;
for l from 0 to m - 3*3 do nops(expand(DD||l)) od;
for l from 0 to m - 4*3 do nops(expand(EE||l)) od;
```

$$l := 0$$
$$\mathbf{1}$$
$$l := 1$$
$$\mathbf{1}$$
$$l := 2$$
$$\mathbf{1}$$
$$l := 3$$
$$\mathbf{1}$$
$$l := 4$$
$$l := 0$$
$$\mathbf{1}$$
$$l := 1$$
$$l := 0$$
$$l := 0$$
$$l := 0 \qquad (13)$$

```
> ## EXPAND ALL UNKNOWNS UP TO y^(13*d)

for l from 0 to m - 0*3 do
AA||l := add(add(A||l[i,j]*x^i*y^j, i=0..dA[l]-j), j=0..min(dA
[l],13*d-1)): od:

for l from 0 to m - 1*3 do
BB||l := add(add(B||l[i,j]*x^i*y^j, i=0..dB[l]-j), j=0..min(dB
[l],13*d-1)): od:

for l from 0 to m - 2*3 do
CC||l := add(add(C||l[i,j]*x^i*y^j, i=0..dC[l]-j), j=0..min(dC
[l],13*d-1)): od:

for l from 0 to m - 3*3 do
DD||l := add(add(D||l[i,j]*x^i*y^j, i=0..dD[l]-j), j=0..min(dD
[l],13*d-1)): od:
```

```
for l from 0 to m - 4*3 do
EE||l := add(add(E||l[i,j]*x^i*y^j, i=0..dE[l]-j), j=0..min(dE
[l],13*d-1)): od:

## SYSTEM FOR DIVISIBILITY BY y^(13*d)

Systemey13d := {}:

for k from 0 to (m-13)/3   do

Coeffs_y13d[13,m-13-3*k,k] := {coeffs(expand(mtaylor(Eq[13,m-13
-3*k,k], y, 13*d)), [x,y])}:

Systemey13d := Systemey13d union Coeffs_y13d[13,m-13-3*k,k]:

od:

Systeme_y13d := Systemey13d:

## VARIABLES FOR DIVISIBILITY BY y^(13*d)

Variables_y13d := indets(Systeme_y13d):

## RESOLUTION y^(13*d)

Solutions_y13d := solve(Systeme_y13d, Variables_y13d):

assign(Solutions_y13d):

##

for l from 0 to m - 0*3 do nops(expand(AA||l)) od;
for l from 0 to m - 1*3 do nops(expand(BB||l)) od;
for l from 0 to m - 2*3 do nops(expand(CC||l)) od;
for l from 0 to m - 3*3 do nops(expand(DD||l)) od;
for l from 0 to m - 4*3 do nops(expand(EE||l)) od;
```

$$l := 0$$
$$\mathbf{1}$$
$$l := 1$$
$$\mathbf{1}$$
$$l := 2$$
$$\mathbf{1}$$
$$l := 3$$
$$\mathbf{1}$$
$$l := 4$$
$$l := 0$$
$$\mathbf{1}$$
$$l := 1$$
$$l := 0$$

$$l := 0$$
$$l := 0 \tag{14}$$

```
> ## EXPAND ALL UNKNOWNS UP TO y^(14*d)

for l from 0 to m - 0*3 do
AA||l := add(add(A||l[i,j]*x^i*y^j, i=0..dA[l]-j), j=0..min(dA
[l],14*d-1)): od:

for l from 0 to m - 1*3 do
BB||l := add(add(B||l[i,j]*x^i*y^j, i=0..dB[l]-j), j=0..min(dB
[l],14*d-1)): od:

for l from 0 to m - 2*3 do
CC||l := add(add(C||l[i,j]*x^i*y^j, i=0..dC[l]-j), j=0..min(dC
[l],14*d-1)): od:

for l from 0 to m - 3*3 do
DD||l := add(add(D||l[i,j]*x^i*y^j, i=0..dD[l]-j), j=0..min(dD
[l],14*d-1)): od:

for l from 0 to m - 4*3 do
EE||l := add(add(E||l[i,j]*x^i*y^j, i=0..dE[l]-j), j=0..min(dE
[l],14*d-1)): od:

## SYSTEM FOR DIVISIBILITY BY y^(14*d)

Systemey14d := {}:

for k from 0 to (m-14)/3   do

Coeffs_y14d[14,m-14-3*k,k] := {coeffs(expand(mtaylor(Eq[14,m-14
-3*k,k], y, 14*d)), [x,y])}:

Systemey14d := Systemey14d union Coeffs_y14d[14,m-14-3*k,k]:

od:

Systeme_y14d := Systemey14d:

## VARIABLES FOR DIVISIBILITY BY y^(14*d)

Variables_y14d := indets(Systeme_y14d):

## RESOLUTION y^(14*d)

Solutions_y14d := solve(Systeme_y14d, Variables_y14d):

assign(Solutions_y14d):

##

for l from 0 to m - 0*3 do nops(expand(AA||l)) od;
```

```
for l from 0 to m - 1*3 do nops(expand(BB||l)) od;
for l from 0 to m - 2*3 do nops(expand(CC||l)) od;
for l from 0 to m - 3*3 do nops(expand(DD||l)) od;
for l from 0 to m - 4*3 do nops(expand(EE||l)) od;
```

$$l := 0$$
$$\mathbf{1}$$
$$l := 1$$
$$\mathbf{1}$$
$$l := 2$$
$$\mathbf{1}$$
$$l := 3$$
$$\mathbf{1}$$
$$l := 4$$
$$l := 0$$
$$\mathbf{1}$$
$$l := 1$$
$$l := 0$$
$$l := 0$$
$$l := 0 \tag{15}$$

```
> ## HOLOMORPHICITY AT INFINITY

for l from 0 to m - 0*3 do

HA||l := add(add(A||l[i,j]*X^i*Y^j*T^(dA[l]-i-j), i=0..dA[l]-j),
j=0..dA[l]):

od:

#

for l from 0 to m - 1*3 do

HB||l := add(add(B||l[i,j]*X^i*Y^j*T^(dB[l]-i-j), i=0..dB[l]-j),
j=0..dB[l]):

od:
```

```
#
for l from 0 to m - 2*3 do
HC||l := add(add(C||l[i,j]*X^i*Y^j*T^(dC[l]-i-j), i=0..dC[l]-j),
j=0..dC[l]):
od:
#
for l from 0 to m - 3*3 do
HD||l := add(add(D||l[i,j]*X^i*Y^j*T^(dD[l]-i-j), i=0..dD[l]-j),
j=0..dD[l]):
od:
#
for l from 0 to m - 4*3 do
HE||l := add(add(E||l[i,j]*X^i*Y^j*T^(dE[l]-i-j), i=0..dE[l]-j),
j=0..dE[l]):
od:
##
for j from 0 to m - 0*3 do
HK[j,0] := HA||j:
od:
#
for j from 0 to m - 1*3 do
HK[j,1] := HB||j:
od:
#
for j from 0 to m - 2*3 do
HK[j,2] := HC||j:
od:
#
for j from 0 to m - 3*3 do
HK[j,3] := HD||j:
```

```
od:
#
for j from 0 to m - 4*3 do
HK[j,4] := HE||j:
od:
```

```
> ##
printlevel := 2:
for q from 0 to m/3   do
for p from 0 to m-3*q do
HEq[p,q] := expand(mtaylor(add(add(add(
  (-1)^k * 2^(p-r) * (d+1)^(j+k-p-q) * binomial(j,r) * (k!)/((p-r)!*q!*(k+r-p-q)!)
* X^(m-q-j-2*k) * T^(k-q) * RY^(2*k-2*q) * HK[j,k],
j=r..m-3*k),
r=max(0,p+q-k)..min(p,m-3*k)),
k=q..m/3)
, T, m-p-3*q)):
od: od;
```

$$q := 0$$
$$HEq := \mathrm{table}([\ ])$$
$$HEq_{0,0} := 0$$
$$HEq_{1,0} := 0$$
$$HEq_{2,0} := 0$$
$$HEq_{3,0} := 0$$
$$q := 1$$
$$HEq_{0,1} := 0$$
$$q := 2 \tag{16}$$

```
> HSysteme := {seq(seq(coeffs(HEq[p,q],[T,X,Y]),p=0..m-3*q),q=0..
  m/3)} minus {0}:

  HSolutions := solve(HSysteme):

  assign(HSolutions):

  ##

  for l from 0 to m - 0*3 do nops(expand(AA||l)) od;
  for l from 0 to m - 1*3 do nops(expand(BB||l)) od;
  for l from 0 to m - 2*3 do nops(expand(CC||l)) od;
  for l from 0 to m - 3*3 do nops(expand(DD||l)) od;
  for l from 0 to m - 4*3 do nops(expand(EE||l)) od;
```

$$l := 0$$
$$\mathbf{1}$$
$$l := 1$$
$$\mathbf{1}$$
$$l := 2$$
$$\mathbf{1}$$
$$l := 3$$
$$\mathbf{1}$$
$$l := 4$$
$$l := 0$$
$$\mathbf{1}$$
$$l := 1$$
$$l := 0$$
$$l := 0$$
$$l := 0 \quad \mathbf{(17)}$$

```
> Jet_xR := factor(

  + (1/Ry^(m-0*2)) * x^(-twist) * add(AA||j * x1^(m-0*3-j) *
  (R1/RR)^j * Delta_xR^0, j=0..m-0*3)

  + (1/Ry^(m-1*2)) * x^(-twist) * add(BB||j * x1^(m-1*3-j) *
  (R1/RR)^j * Delta_xR^1, j=0..m-1*3)

  + (1/Ry^(m-2*2)) * x^(-twist) * add(CC||j * x1^(m-2*3-j) *
  (R1/RR)^j * Delta_xR^2, j=0..m-2*3)

  + (1/Ry^(m-3*2)) * x^(-twist) * add(DD||j * x1^(m-3*3-j) *
  (R1/RR)^j * Delta_xR^3, j=0..m-3*3)

  + (1/Ry^(m-4*2)) * x^(-twist) * add(EE||j * x1^(m-4*3-j) *
  (R1/RR)^j * Delta_xR^4, j=0..m-4*3) );
```

$$Jet\_xR := 0 \tag{18}$$

```
> Vars := indets(Jet_xR) minus {x, x1, x2, y, R1, R2};
```

```
NbVars := nops(Vars);
```

$$Vars := :$$
$$NbVars := 0 \tag{19}$$

```
> Ens_Vars := {}:
for i from 1 to nops(Vars) do
Ens_Vars := Ens_Vars union {op(i,Vars)=0}:
od:
EnsVars := Ens_Vars;
```

$$EnsVars := : \tag{20}$$

```
> for i from 1 to nops(Vars) do
JetSol[i] := factor(subs(EnsVars, subs(op(i,Vars)=1, Jet_xR))):
od;
```

$$i := 1 \tag{21}$$